

So the water level doesn't change.

2(a) Work done is defined as the product of force and the distance moved in the direction of the force.

SI unit: J, joules

(b)

Situation	Numerical value of the work that can be done	On what object is the work done?	Application of conservation of energy
Hot air balloon mass 800 kg landing from an altitude of 1500m moving at constant speed	GPE = mgh = (800)(9.81)(1500) = 1.2×10^7 J	Work is done by the balloon on the air below and surrounding it.	Loss in gravitational potential energy of balloon is converted to the gain in kinetic energy of air around the balloon.
Battery e.m.f 12.0V pushing a total charge of 800 C through an electric drill	$V = \frac{W}{Q}$ $W = VQ = 12 \times 800$ $W = 9600$ J	Work is done on the drill	Electrical energy is converted to kinetic energy of drill and also sound and heat energy.